

CASE STUDY · THE METHOD, MADE FALSIFIABLE

How a concept dies

The argument is that the pipeline is auditable, not that it is clever. So here is one idea walked through the whole gauntlet — and published the moment it lost. No averted look.

Every concept on this site follows the same path. The survivors you can buy, the free sample, and the 55 rejections in the graveyard all started the same place: a paper reporting an effect that looks like it ought to be a business. What separates them is which stage killed the ones that died. Below, two that did.

The process, in one line each

- **Discovery.** A peer-reviewed mechanism that is extensively documented and that nobody is selling.
- **Market-absence search.** Suppliers, patents, filings, industry press — reported with counts and named near-substitutes, so "absent" is a search result, not an assertion.
- **Economic model.** The cost model built from the paper's own primitives (feedstock, yield, batch time, energy) and recomputed in code.
- **Gate scoring.** Seven gates, no averaging, no partial credit: scientific backing, commercial absence, unit economics, barrier to entry, velocity, market fit, demonstrated superiority.
- **Red-team audit.** A separate, deterministic step that argues against our own work — including against our survivors.

- **Two-part superiority verdict.** Scientific superiority and commercial superiority are judged separately, and never averaged.
- **Decision.** Published either way — the surviving dossier, or the autopsy and the gate that killed it.

Walkthrough: the concept that beat the science and lost anyway

High-yield biomanufactured magnetosomes (FIONs) for ultra-high relaxivity MRI

contrast. This one is a clean illustration of the single most common failure: the science is real, and the business was never anchored to a product that exists.

Discovery. The literature reported magnetosome iron-oxide nanoparticles with an r_2 relaxivity of **324–471 mM⁻¹s⁻¹** against **133 mM⁻¹s⁻¹** for Feridex — roughly a **2.4–3.5×** improvement. On a raw lab reading, that is a step-change in contrast efficacy.

Market-absence search. The market-absence ledger is where it fell. The comparator used for that head-to-head, **Feridex, was withdrawn from the market in 2008.** It is not absent because nobody noticed it; it is absent because it was discontinued. The real current comparators for MRI contrast — the products a buyer actually chooses among today — are a different family, at different prices.

Economic model. Because the comparison was anchored to a discontinued product, the margin model was priced against **Feridex's historical price**, not the price of anything a buyer can buy. The gross margin and payback were therefore unestablished — the model was built on a number nobody can transact at.

Velocity and regulatory path. MRI contrast is a regulated medical product. The route was priced at **\$35M and 84 months** to first revenue, with high regulatory risk. Against our velocity cap, that is not a venture — it is a decade-long clinical program.

The two-part verdict. *Scientific superiority: yes.* The mechanism genuinely outperforms its stated comparator. *Commercial superiority: no.* The comparator was dead, the price anchor was historical fiction, and the path to a paid delivery was prohibitively long and expensive. A real improvement over a product nobody sells is not a business.

Decision: rejected. Voided on incumbent legitimacy, with economics that could not be established. Specific failure modes recorded in the ledger: dead incumbent, economics undisclosed margin, capital and time to revenue.

Contrast: the concept that beat the wrong thing

Plant-derived urease-driven enzyme-induced carbonate precipitation (EICP). Same trap, different shape — the comparator was a straw man rather than a corpse.

The scientific claim. The reported effect was a **1.83×** improvement in unconfined compressive strength — **6.41 MPa** against a **3.5 MPa** result for a **10% Portland-cement mix**, at a raw-material cost quoted at **3.3×** higher (**\$52 /m³** against **\$15.52 /m³**).

Why it failed. The 10% mix is not the product a buyer uses. Ordinary Portland cement sits in a range of roughly **1.5–8 MPa**; **6.41 MPa sits inside that ordinary range**. Beating a deliberately weak laboratory blend proves only that the method beats that blend. Against the real market-leading product, it is not better — and it is **3.3×** the material cost to be merely comparable. It is more expensive to be equal.

The two-part verdict. *Scientific superiority: not demonstrated against the real incumbent. Commercial superiority: no* — at a higher cost for an equal result, there is no switching incentive. **Decision: rejected** — voided for a straw incumbent and for being commercially weaker.

Why we show you the bodies

If a concept dies, publishing how it died is the honest move — and it is what makes the survivors believable. The same pipeline that produced these two rejections cleared the dossiers on sale. When you read the free sample, you are seeing the output of a machine that also publishes its own failures, *inside* the document.

A rejection is not a claim that the science is wrong. It is a claim that the venture case did not survive: the comparator was not what a buyer actually buys, the comparison was not like-for-like, the advantage was not a step change, or the capital and time to first revenue were prohibitive.

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